

7 (a) 92 protons and 143 neutrons B1 [1]

(b)

	value
a	1
b	0
c	141
d	55

(a and b both required)

B1
B1
B1 [3]

(c) kinetic energy (of products) or gamma/ γ (radiation or photon) B1 [1]

(d) (total) mass on left-hand side/reactants is greater than (total) mass on right-hand side/products

M1

difference in mass is (converted to) energy

A1 [2]